Practically Implementable Minimal Universal Gate Sets for Multi-Qudit Systems with Cryptographic Validation

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Journey of Quantum Cryptography

Timeline of Quantum Computing Challenging Cryptographic Hardness

Foundations of Modern Cryptology

Classical hardness, cryptographic primitives, provable security.

Diffie-Hellman ’76
DES ’77
RSA ’77
ECC ’85
RC4 ’87
SHA-1 ’95
AES ’01

1970s – 2000s

Quantum Emergence

Quantum algorithms, quantum threats, disruption of classical security.

BB84 ’84
Quantum Teleportation ’93
Shor ’94
Grover ’96

1990s – 2010s

Theorization of Qudits

Higher-dimensional states, generalized operators, expanded quantum models.

Generalized Qudit Pauli ’10
Reck’s Method Generalization ’10
d-Dimensional Entanglement ’12

2010s – 2020s

Qudit Supremacy in Cryptology

High-dimensional protocols, enhanced security, advanced entanglement techniques.

Qudit-based secret-sharing ’20
Quantum fingerprinting ’22
Qudit RS & Qudit CSS ’23

Present Days

Physical Qudit Cryptosystem

Multi-level Qudits, practical hardware platforms, scalable implementations.

Superconducting Qutrit Gates ’19
Trapped-ion Qudits ’22
Photonic Qudit Chips ’24

Present Days

Motivation

Literature Survey and Key Contributions of the Paper

Paper Title

Practically Implementable Minimal Universal Gate Sets
for Multi-Qudit Systems with Cryptographic Validation

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Targeted Advancements over the Prior Works

Decomposition Method	Applicability	Universality	Minimality	Implementable
Barenco et al. (1995)	**Qubit** Systems	☑	☒	☑
Solovay Kitaev (1997)	**Qubit** Systems	☑	☑	☑
Reck et al. (1994)	**Multi-Qudit** Systems	☑	☒	☑
Jarlskog (2005)	**Multi-Qudit** Systems	☑	☒	☒
Li Yin Roberts (2012)	**Multi-Qudit** Systems	☑	☒	☑
Our Proposed Method	**Multi-Qudit** Systems	☑	☑	☑

Problem Statement

Formal Research Proposal of the Paper

$$\mathcal{S} = T_{elements} \cup PHASE_1$$

The set $\mathcal{S}$ serves as a minimal set of universal gates for multi-dimensional Qudits of N folds.

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$$U = \begin{bmatrix} u_{11} & u_{12} \\ u_{21} & u_{22} \end{bmatrix} \in SU(2) \quad \ggg \quad T_{i,j}(U) = \begin{bmatrix} 1\;0 & & \cdots & & 0\;0 \\ 0\;1 & & \cdots & & 0\;0 \\ \vdots\;\vdots & \ddots & & & \vdots\;\vdots \\ 0\;0 & u_{11} & \cdots & u_{12} & 0\;0 \\ \vdots\;\vdots & \vdots & \cdots & \vdots & \vdots\;\vdots \\ 0\;0 & u_{21} & \cdots & u_{22} & 0\;0 \\ \vdots\;\vdots & & & \ddots & \vdots\;\vdots \\ 0\;0 & & \cdots & & 1\;\;0 \\ 0\;\;0 & & \cdots & & 0\;\;1 \end{bmatrix} \in SU(N^n)$$

$$\boldsymbol{T_{elements}} = \left\{ R_j = \prod T_{i,j}(U) \,\middle|\, 0 \leq i < \; j \leq N^n, U \in SU(2) \right\} \subset SU(N^n)$$

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$$Phase(j, \Phi_j) = \begin{bmatrix} 1\;0 & & \cdots & & 0\;0 \\ 0\;1 & & \cdots & & 0\;0 \\ \vdots\;\vdots & \ddots & & & \vdots\;\vdots \\ 0\;0 & 1 & 0 & 0 & 0\;0 \\ 0\;0 & 0 & e^{i\Phi_j} & 0 & 0\;0 \\ 0\;0 & 0 & 0 & 1 & 0\;0 \\ \vdots\;\vdots & & & \ddots & \vdots\;\vdots \\ 0\;0 & & \cdots & & 1\;0 \\ 0\;0 & & \cdots & & 0\;1 \end{bmatrix}$$

$$\boldsymbol{PHASE} = \{Phase(j, \Phi_j) \mid j \in \{0, \dots, N^n - 1\}, \Phi_j \in [-\pi, \pi)\}$$

$$\boldsymbol{PHASE_1} = \{Phase(1, \theta) \mid \theta \in [-\pi, \pi)\} \subset PHASE$$

Reck's Decomposition

Building Block of our Proposed Decomposition Method

Reck's Decomposition Method

$\boldsymbol{S'} = T_{elements} \cup \boldsymbol{PHASE}$

The set $\boldsymbol{S}'$ serves as universal gate set
for multi-dimensional Qudits of $\boldsymbol{N}$ folds.

Any unitary matrix can be decomposed into **two-dimensional subspace rotations** and a diagonal **phase shift.**

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$$\boldsymbol{M_4} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \rightarrow \boldsymbol{M_4 T_{4,3}} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & 0 \\ * & * & 0 & e^{i\theta_j} \end{bmatrix} \rightarrow \boldsymbol{M_4\, T_{4,3} T_{4,2} T_{4,1}} = \begin{bmatrix} * & * & * & 0 \\ * & * & * & 0 \\ * & * & * & 0 \\ 0 & 0 & 0 & e^{i\theta_j} \end{bmatrix} = \mathbf{M_4 R_4}$$

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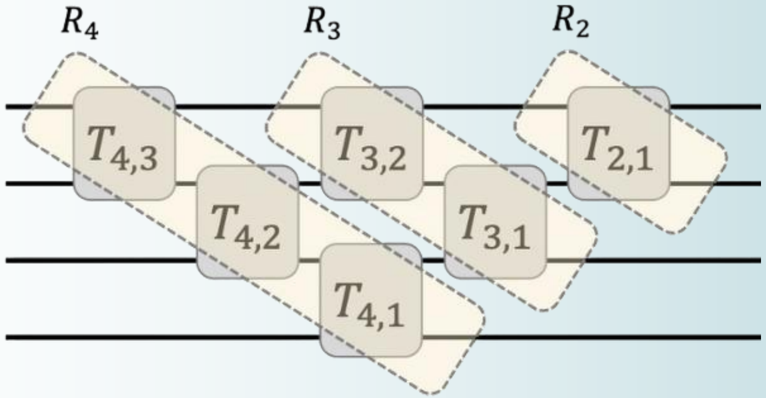

$$M_4 = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & * \\ * & * & * & * \end{bmatrix} \rightarrow M_4 T_{4,3} = \begin{bmatrix} * & * & * & * \\ * & * & * & * \\ * & * & * & 0 \\ * & * & 0 & e^{i\theta_j} \end{bmatrix} \rightarrow M_4\ T_{4,3} T_{4,2} T_{4,1} = \begin{bmatrix} * & * & * & 0 \\ * & * & * & 0 \\ * & * & * & 0 \\ 0 & 0 & 0 & e^{i\theta_j} \end{bmatrix} = \mathbf{M_4 R_4}$$

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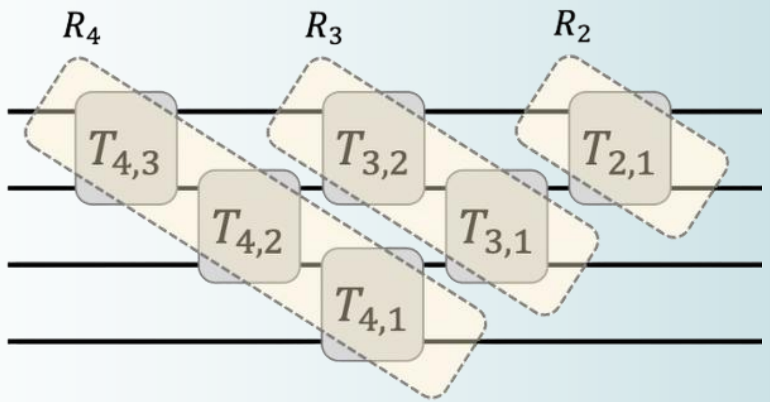

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$$\boldsymbol{M_4 R_4 R_3 R_2} = \begin{bmatrix} e^{i_1\theta_{j_1}} & 0 & 0 & 0 \\ 0 & e^{i_2\theta_{j_2}} & 0 & 0 \\ 0 & 0 & e^{i_3\theta_{j_3}} & 0 \\ 0 & 0 & 0 & e^{i_4\theta_{j_4}} \end{bmatrix} = \boldsymbol{\Phi} \in PHASE \quad \rightarrow \quad \boldsymbol{M_4} = \prod \boldsymbol{R_k\, \Phi}$$

Any unitary matrix can be decomposed into **two-dimensional subspace rotations** and a diagonal **phase shift.**

Proof of Sufficiency

Establishing Constructional Universality of our Proposed Decomposition Method

Reck's Decomposition Method

$\boldsymbol{S}' = T_{elements} \cup \boldsymbol{PHASE}$

The set $\boldsymbol{S}'$ serves as universal gate set for multi-dimensional Qudits of $\boldsymbol{N}$ folds.

Our Proposed Method

$\boldsymbol{S} = T_{elements} \cup \boldsymbol{PHASE_1}$

The set $\boldsymbol{S} \subset \boldsymbol{S}'$ serves as a minimal set of universal gates for multi-dimensional Qudits of $\boldsymbol{N}$ folds.

Any diagonal phase shift matrix can be decomposed into **two-dimensional subspace rotations** and a **phase1 matrix.**

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$$\boldsymbol{\Phi} = \begin{bmatrix} e^{i\theta_0} & 0 & \cdots & 0 \\ 0 & e^{i\theta_1} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & e^{i\theta_k} \end{bmatrix} = \begin{bmatrix} e^{i\theta_0} & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & e^{i\theta_1} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} \cdots \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & e^{i\theta_k} \end{bmatrix} = \prod_j Phase(j, \Phi_j)$$

Any diagonal phase shift matrix can be decomposed into **two-dimensional subspace rotations** and a **phase1 matrix.**

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Any diagonal phase shift matrix can be decomposed into **two-dimensional subspace rotations** and a **phase1 matrix.**

Proof of Necessity

Establishing Functional Minimality of our Proposed Decomposition Method

Reck's Decomposition Method

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The set $\mathcal{S}'$ serves as universal gate set for multi-dimensional Qudits of $\boldsymbol{N}$ folds.

Our Proposed Method

$\mathcal{S} = T_{elements} \cup \boldsymbol{PHASE_1}$

The set $\mathcal{S} \subset \mathcal{S}'$ serves as a **minimal** set of universal gates for multi-dimensional Qudits of $\boldsymbol{N}$ folds.

Applicability	$T_{elements}$	$\boldsymbol{PHASE_1}$
Mixing Rotation between Basis States	✓	✗
(Non-trivial) Phase Freedom	✗	✓
Simulating each other Set	✗	✗

$\boldsymbol{T_{elements}}$ is required for **off-diagonal mixing** and $\boldsymbol{PHASE_1}$ is necessary for **Global Phase Generation.**

Functional Architecture

Structural Flow of the Proof as Followed in the Paper

Reck's Decomposition Method

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Constructing the Building Blocks

- Construction of $R_{i,j}$ from $T_{elements}$ set
- +
- Construction of ϕ from $PHASE_1$ set

Connection to Reck's decomposition

- Reck uses only $R_{i,j}$ and ϕ for decomposition
- All matrices in Reck's decomposition can be constructed with $\mathcal{S}$

Universality & Minimality

- Any Multi-Qudit gate can be constructed with $\mathcal{S}$
- +
- Minimality of $\mathcal{S}$ wrt Generator Minimality

$\mathcal{S}$ is Minimal Universal Multi-Qudit Gate Set

Decomposition Algorithm

Initialization

Input : A multi-dimensional multi-fold Qudit gate M of order N

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Constructing 2-level Rotations

Embed a tiny 2*2 rotation U to $\mathrm{T}_{i,j}$ (on $|i\rangle, |j\rangle$ components) into the full Qudit space

Decomposition Algorithm

Iterate i, j over N

Continue until all sub-diagonal elements are eliminated

Initialization → Constructing 2-level Rotations → Column wise reduction

Input : A multi-dimensional multi-fold Qudit gate M of order N

Embed a tiny 2*2 rotation U to $\mathrm{T}_{i,\mathrm{j}}$ (on $|\mathrm{i}\rangle, |\mathrm{j}\rangle$ components) into the full Qudit space

Sub-diagonal i, j-th entries of M to be *cancelled* using the constructed 2-level Rotation operations $R_{i,\mathrm{j}}$, yielding Diagonal $PHASE$ matrix

$$\boldsymbol{M} = \prod_{\substack{\boldsymbol{i},\mathrm{j} = 0 \to \mathrm{N}-1 \\ \mathrm{i} < \mathrm{j}}} R_{i,\mathrm{j}}\ \Phi$$

$R_{i,\mathrm{j}} \in \mathrm{T_{elements}} = \{ \prod \mathrm{T}_{i,\mathrm{j}}(\mathrm{U}) \mid \mathrm{U} \in \mathrm{SU}(2)\}$, Φ Diagonal Phase matrix

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Decompose the diagonal unitary

Diagonal $PHASE$ matrix to be decomposed into $\mathrm{T}_{\mathrm{elements}}$ and $PHASE_1$ gates.

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$$\Phi = \prod_{\substack{\mathrm{k} = 0 \to \mathrm{n} \\ \mathrm{n} \leq \mathrm{N}-1}} T_{\mathrm{k}}\ \Phi'$$

$T_{\mathrm{k}} \in \mathrm{T}_{\mathrm{elements}}$, Φ' Phase-1 matrix

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$$\boldsymbol{M} = \prod_{\substack{i,j = 0 \to N-1 \\ i<j}} R_{i,j}\, \Phi$$

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Decompose the diagonal unitary

Diagonal $PHASE$ matrix to be decomposed into $T_{elements}$ and $PHASE_1$ gates.

$$\Phi = \prod_{\substack{k = 0 \to n \\ n \le N-1}} T_k\, \Phi'$$

$T_k \in T_{elements}$, Φ' Phase-1 matrix

Structure the decomposition

Diagonal $PHASE$ matrix to be decomposed into $T_{elements}$ and $PHASE_1$ gates.

Decomposition Algorithm Visualization

Iterate i, j over $N = 3$

Continue until all sub-diagonal elements are eliminated

Initialization → Constructing 2-level Rotations → Column wise reduction → Decompose the diagonal unitary → Structure the decomposition

$$M = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}$$

Hadamard Qutrit Gate

$$R_{0,2}^{-1} = \begin{pmatrix} 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ 0 & 1 & 0 \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{pmatrix}$$

$$R_{0,1}^{-1} = \begin{pmatrix} 0.81 & 0.57 & 0 \\ -0.57 & 0.81 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$R_{1,2}^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -0.4-0.6i & -0.4+0.61i \\ 0 & 0.4+0.6i & -0.4-0.61i \end{pmatrix}$$

$$\Phi = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -0.5+0.9i & 0 \\ 0 & 0 & -0.5-0.9i \end{pmatrix}$$

$$R_{0,2}^{-1}\, M = U^{(1)},\; R_{0,1}^{-1}\, U^{(1)} = U^{(2)},\; R_{1,2}^{-1}\, U^{(2)} = U^{(3)} \rightarrow \boldsymbol{M = R_{0,2}\; R_{0,1}\; R_{1,2}\; \Phi}$$

$$\boldsymbol{\Phi = T_1 * T_2 * I_3}$$

$$T_1 = \begin{pmatrix} -0.5-0.9i & 0 & 0 \\ 0 & -0.5+0.9i & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$T_2 = \begin{pmatrix} -0.5+0.9i & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -0.5-0.9i \end{pmatrix}$$

$$P_3 \simeq \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\boldsymbol{M = R_{0,2,} R_{0,1} R_{1,2} T_1 T_2 P_3}$$

$R_{0,2}, R_{0,1}, R_{1,2}, T_1, T_2 \in \mathrm{T}_{\mathrm{elements}}$

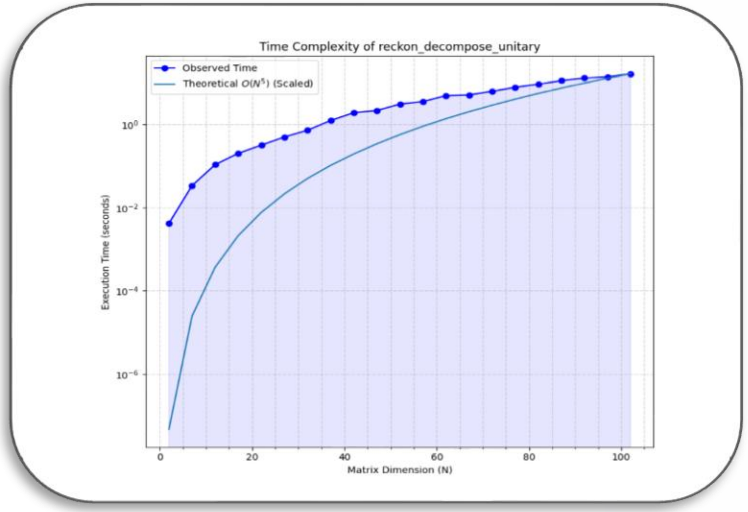
$\boldsymbol{P_3} \in PHASE_1$

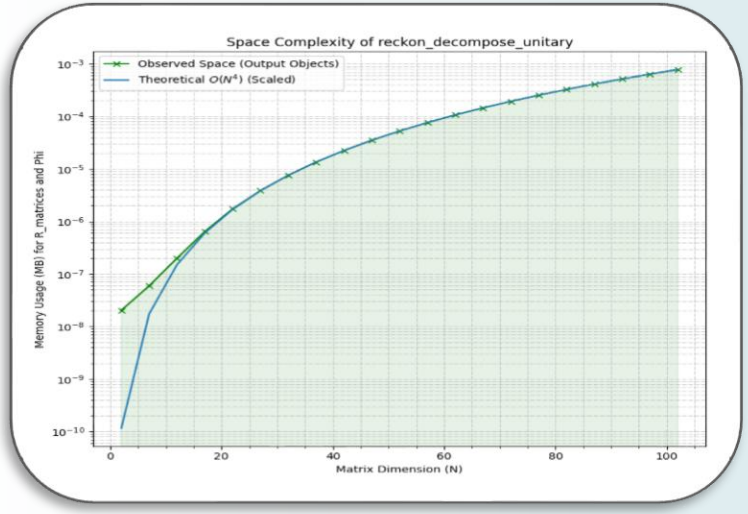
Algorithmic Performance Overview

Complexity Bounds and Accuracy Metrics

Polynomial Time Complexity

$$O((N^n)^5)$$



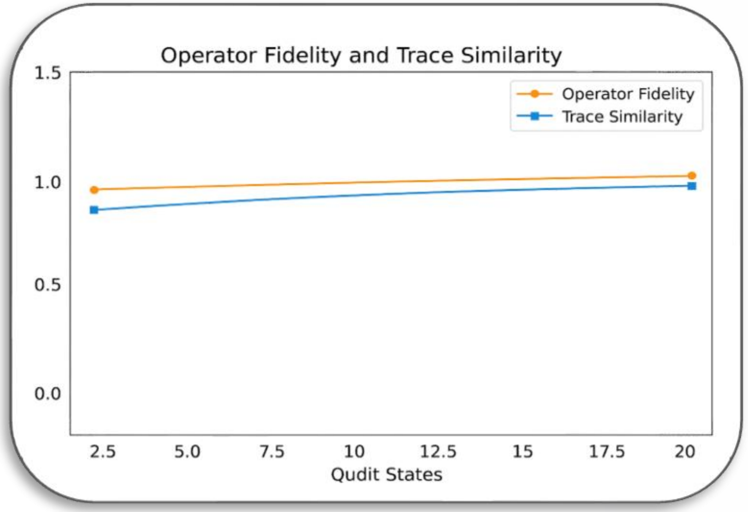


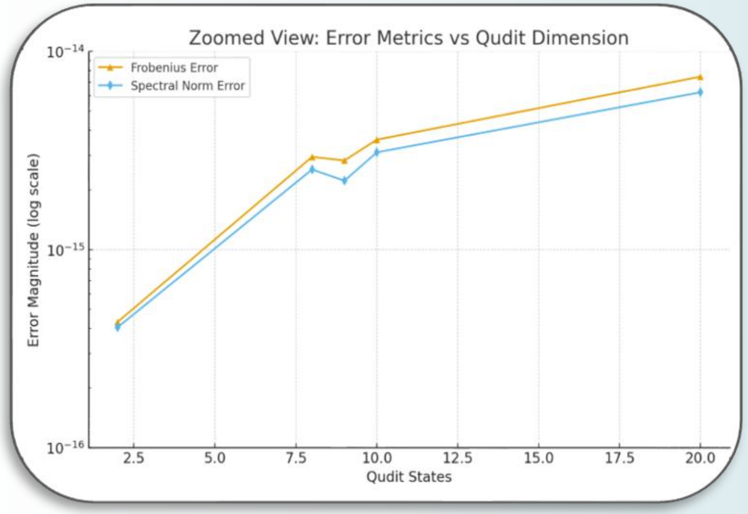
Polynomial Space Complexity

$$O((N^n)^4)$$

High Operator Fidelity and Trace Similarity

$$1 \pm O(10^{-16})$$



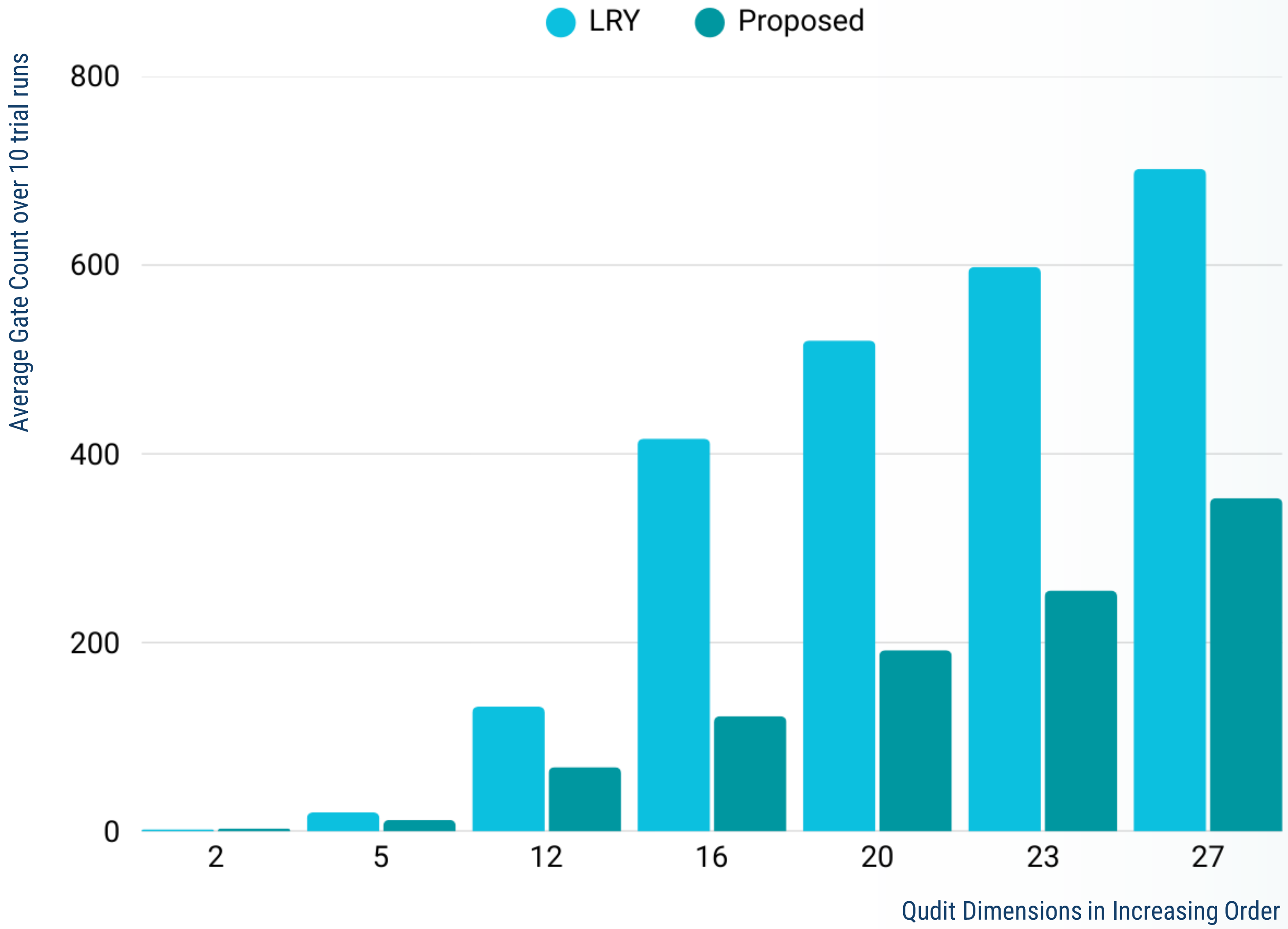


Low Reconstruction Errors
Frobenius Error
Spectral Norm Error

$$O(10^{-16})$$

Performance Comparison

Gate Complexity Comparison with Existent Established Method



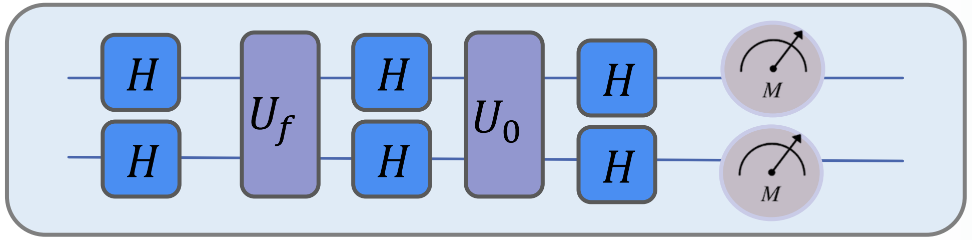
Advantages over
Li-Roberts-Yin's (LRY)
Decomposition Method Unitary Matrices and Quantum Gates **(2013)**

- Generator Minimality
- Lower Computation Overhead
- Higher Reconstruction Accuracy

Cryptographic Algorithm Simulation

Simulation of Grover's Algorithm using Traditional and Decomposed Ququart Gates (dim 4)

Grover's Algorithm Architecture with Classical Gates



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Simulation of Grover's Algorithm using Traditional and Decomposed Ququart Gates (dim 4)

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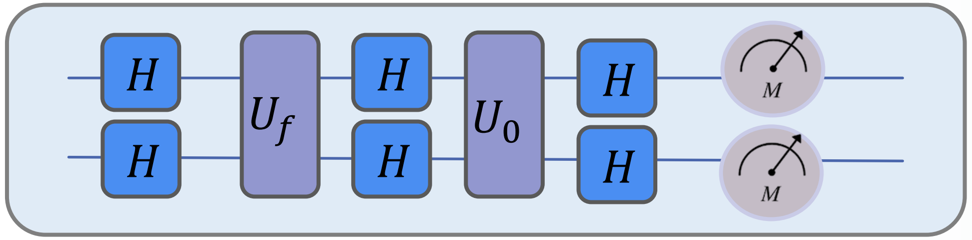
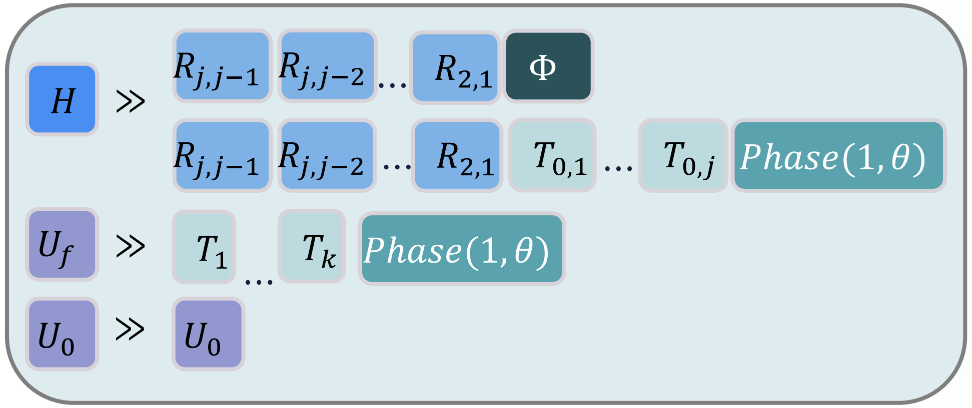


Decomposition of Hadamard Gate, U_f Gate & U_0 Gate



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Simulation of Grover's Algorithm using Traditional and Decomposed Ququart Gates (dim 4)

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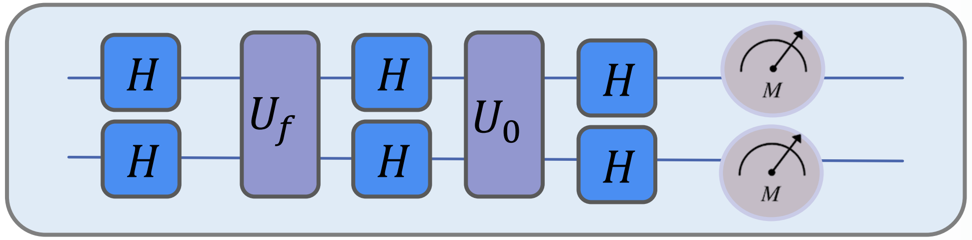
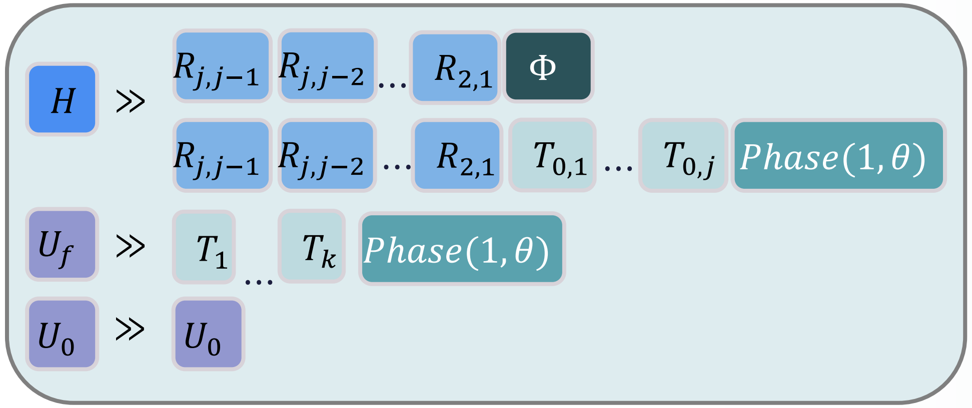
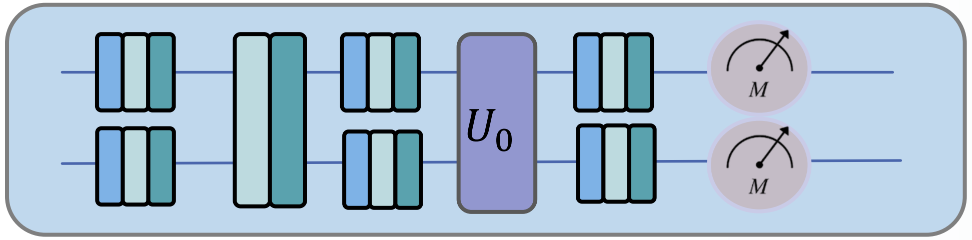


Decomposition of Hadamard Gate, U_f Gate & U_0 Gate



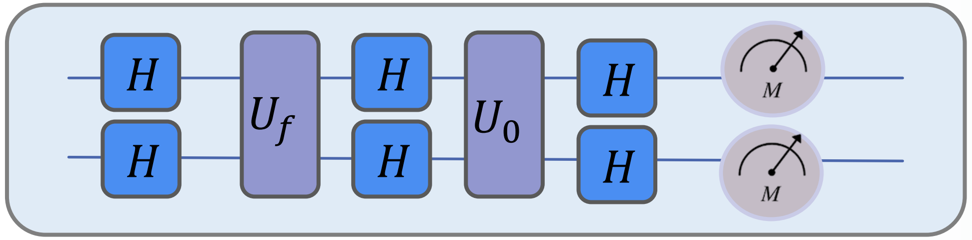
Grover's Algorithm Architecture with Decomposed Gates



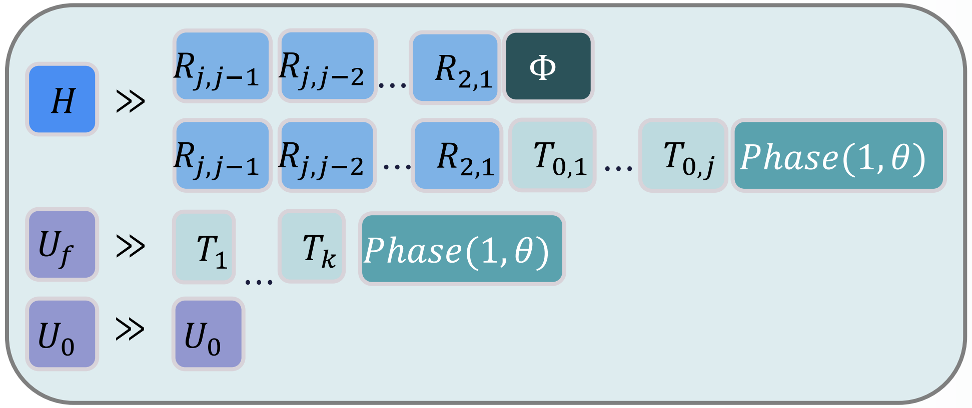
Cryptographic Algorithm Simulation

Simulation of Grover's Algorithm using Traditional and Decomposed Ququart Gates (dim 4)

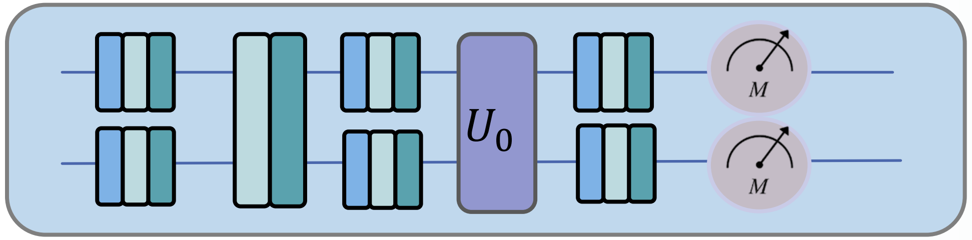
Grover's Algorithm Architecture with Classical Gates

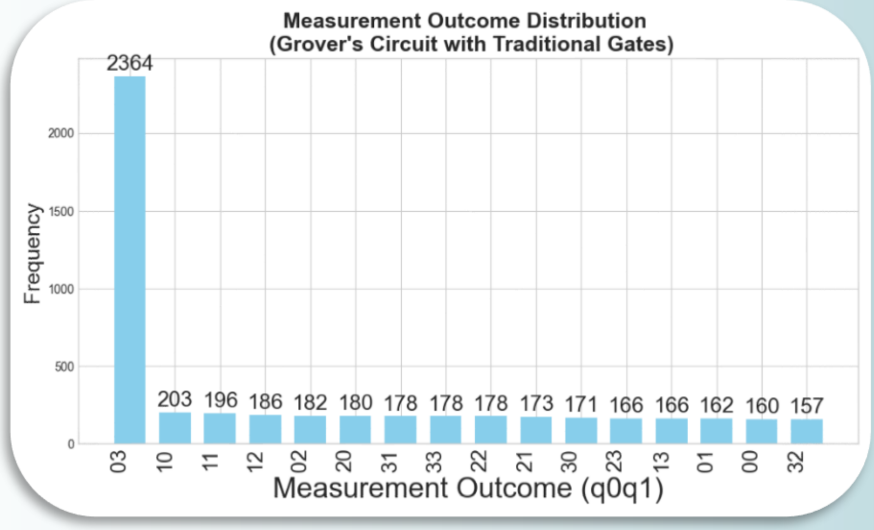


Decomposition of Hadamard Gate, U_f Gate & U_0 Gate

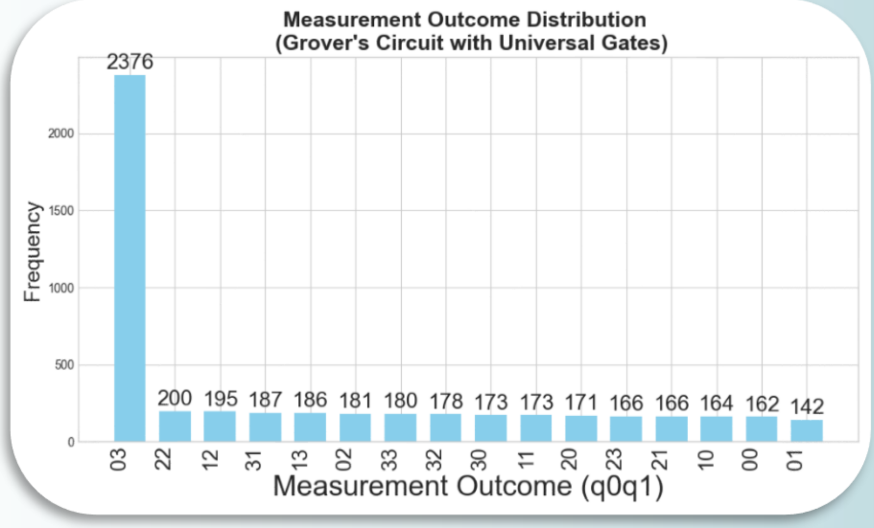


Grover's Algorithm Architecture with Decomposed Gates





Grover's Algorithm Simulation for 4-dim Qudit With '03' marked state



Cryptographic Algorithm Simulation

Simulation of Quantum key Distribution using Traditional and Decomposed Qutrit Gates (dim 3)

Choice of Basis	Choice of Qutrit	Contribution in Circuit
Rectilinear	0	
Rectilinear	1	$0S1$
Rectilinear	2	$0S2$
Diagonal	0	H
Diagonal	1	$0S1$ — H
Diagonal	2	$0S2$ — H

Alice

Bob

Choice of Basis	Contribution in Circuit
Rectilinear	R
Diagonal	H — D

Cryptographic Algorithm Simulation

Simulation of Quantum key Distribution using Traditional and Decomposed Qutrit Gates (dim 3)

Choice of Basis	Choice of Qutrit	Contribution in Circuit
Rectilinear	0	
Rectilinear	1	$0S1$
Rectilinear	2	$0S2$
Diagonal	0	H
Diagonal	1	$0S1$ — H
Diagonal	2	$0S2$ — H

Alice

Bob

Choice of Basis	Contribution in Circuit
Rectilinear	R
Diagonal	H — D

Decomposition of Hadamard, $0swap1$ and $0swap2$ Gates

$H \gg R_3 \; R_2 \; T_2 \; T_1 \; Phase(1,\theta)$

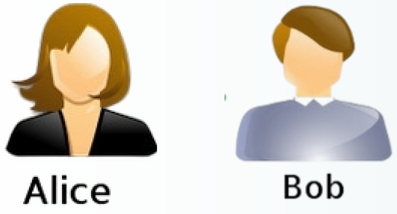
$0S1 \gg R'_1 \; R'_0 \; Phase(1,\theta_1)$

$0S2 \gg R''_2 \; R''_0 \; Phase(1,\theta_2)$

Cryptographic Algorithm Simulation

Simulation of Quantum key Distribution using Traditional and Decomposed Qutrit Gates (dim 3)

Choice of Basis	Choice of Qutrit	Contribution in Circuit
Rectilinear	0	—
Rectilinear	1	$0S1$
Rectilinear	2	$0S2$
Diagonal	0	H
Diagonal	1	$0S1$ — H
Diagonal	2	$0S2$ — H



Choice of Basis	Contribution in Circuit
Rectilinear	R (measurement)
Diagonal	H — D (measurement)

Decomposition of Hadamard, $0swap1$ and $0swap2$ Gates

$H \gg R_3\ R_2\ T_2\ T_1\ Phase(1,\theta)$

$0S1 \gg R'_1\ R'_0\ Phase(1,\theta_1)$

$0S2 \gg R''_2\ R''_0\ Phase(1,\theta_2)$

Chosen Qutrit **0**
Chosen Basis **Rectilinear**

Chosen Basis **Rectilinear**

R — Traditional Circuit

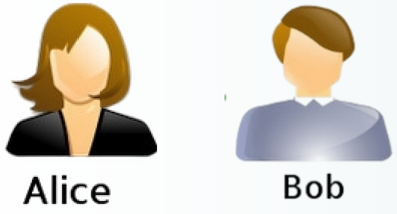
R — Circuit with Decomposed Gates

Alice and Bob used same bases.
Key Qutrit established for this round 0

Cryptographic Algorithm Simulation

Simulation of Quantum key Distribution using Traditional and Decomposed Qutrit Gates (dim 3)

Choice of Basis	Choice of Qutrit	Contribution in Circuit
Rectilinear	0	
Rectilinear	1	$0S1$
Rectilinear	2	$0S2$
Diagonal	0	H
Diagonal	1	$0S1$ — H
Diagonal	2	$0S2$ — H



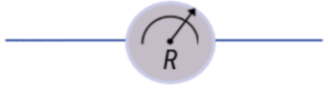
Choice of Basis	Contribution in Circuit
Rectilinear	R
Diagonal	H — D

Decomposition of Hadamard, $0swap1$ and $0swap2$ Gates

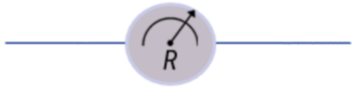
H	$\gg$	R_3	R_2	T_2	T_1	$Phase(1, \theta)$
$0S1$	$\gg$	R'_1	R'_0	$Phase(1, \theta_1)$		
$0S2$	$\gg$	R''_2	R''_0	$Phase(1, \theta_2)$		

Chosen Qutrit **0**
Chosen Basis **Rectilinear**

Chosen Basis **Rectilinear**



Traditional Circuit

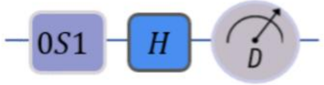


Circuit with Decomposed Gates

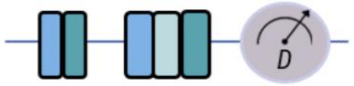
Alice and Bob used same bases.
Key Qutrit established for this round 0

Chosen Qutrit **1**
Chosen Basis **Rectilinear**

Chosen Basis **Diagonal**



Traditional Circuit



Circuit with Decomposed Gates

Alice and Bob used different bases.
No key Qutrit established for this round.

Cryptographic Algorithm Simulation

Simulation of Quantum key Distribution using Traditional and Decomposed Qutrit Gates (dim 3)

Choice of Basis	Choice of Qutrit	Contribution in Circuit
Rectilinear	0	
Rectilinear	1	$0S1$
Rectilinear	2	$0S2$
Diagonal	0	H
Diagonal	1	$0S1$ — H
Diagonal	2	$0S2$ — H

Alice

Bob

Choice of Basis	Contribution in Circuit
Rectilinear	R
Diagonal	H — D

Decomposition of Hadamard, $0swap1$ and $0swap2$ Gates

H	$\gg$	R_3	R_2	T_2	T_1	$Phase(1,\theta)$
$0S1$	$\gg$	R'_1	R'_0	$Phase(1,\theta_1)$		
$0S2$	$\gg$	R''_2	R''_0	$Phase(1,\theta_2)$		

Alice
Chosen Qutrit **0**
Chosen Basis **Rectilinear**

Bob
Chosen Basis **Rectilinear**

R Traditional Circuit

R Circuit with Decomposed Gates

Alice and Bob used same bases.
Key Qutrit established for this round 0

Alice
Chosen Qutrit **1**
Chosen Basis **Rectilinear**

Bob
Chosen Basis **Diagonal**

$0S1$ — H — D Traditional Circuit

D Circuit with Decomposed Gates

Alice and Bob used different bases.
No key Qutrit established for this round.

Alice
Chosen Qutrit **2**
Chosen Basis **Diagonal**

Bob
Chosen Basis **Diagonal**

$0S2$ — H — H — D Traditional Circuit

D Circuit with Decomposed Gates

Alice and Bob used same bases.
Key Qutrit established for this round 2

End of Presentation.

Questions Please.

Indocrypt 2025
IIIT Bhubaneswar
December 15th 2025

GitHub Code Repository